

CHARTING THE HEAVENS

PEOPLE HAVE OBSERVED THE HEAVENS FOR THOUSANDS OF YEARS, AND MANY CULTURES HAVE LINKED THE PATTERNS THEY DISCERNED AMONG THE THOUSANDS OF VISIBLE STARS TO THEIR OWN MYTHOLOGY.

Today, the International Astronomical Union (IAU) recognizes 88 constellations and together the constellations form a complete sphere (see pp.94–95) around the Earth. Our modern system of constellations is based on the 48 figures described by the ancient Greek astronomer Ptolemy. Other civilizations also visualized patterns in the sky and linked those to their myths and legends, but only the Greek system is recognized today. It was not until the 16th century, when sailors started to navigate and explore the Southern Hemisphere, that whole new areas of the celestial sphere were mapped and new constellations created.

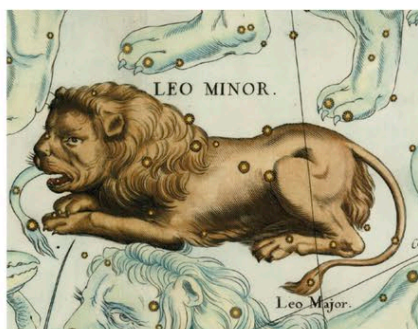


Edmond Halley

1679

Halley's southern survey

English astronomer Edmond Halley makes the first accurate survey of the southern sky from the island of St. Helena. His catalog contains 341 stars, and he introduces a new constellation, Robur Carolinum (Charles's Oak), but it is not accepted by other astronomers.



Hevelius's Leo Minor

1690

Hevelius's new constellations

Johannes Hevelius, a Polish astronomer, publishes a catalog of more than 1,500 stars, larger and more accurate than that of Tycho Brahe, along with a new star atlas. Hevelius introduces ten new constellations, seven of which are still accepted by astronomers today.

Babylonian clay tablet



3000–1000 BCE

Dawn of astronomy

Sumerian and Babylonian astronomers watch the yearly motions of the Sun and stars and create the first constellations, such as GUD.AN.NA, the modern Taurus. Their observations are recorded in cuneiform script on clay tablets like this one.

Globe with early Greek constellations

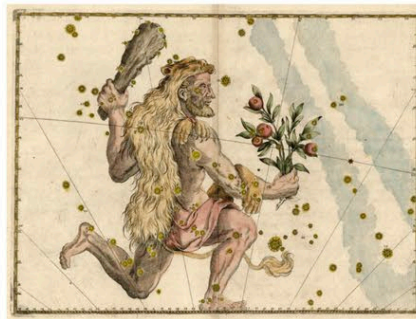


400–250 BCE

First Greek constellation system

Eudoxus, a Greek astronomer, introduces Babylonian constellations to the West in amended form in a book entitled *Phaenomena*. His original text is long lost, but it was turned into an instructional poem by another Greek, Aratus, and later translated into Latin.

Hercules by Bayer



1603

First all-sky star atlas

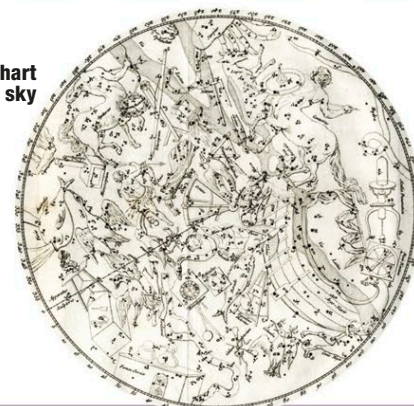
Johann Bayer, a German lawyer and amateur astronomer, publishes the first celestial atlas to cover the whole sky, *Uranometria*. He assigns a full page to each of the 48 Ptolemaic constellations, with an additional page for the 12 new southern constellations.

1592–1612

New constellations

Petrus Plancius, a Dutch cartographer and astronomer, introduces 15 new constellations. Twelve of them lie in the far southern sky that is invisible from Europe and include stars plotted by Dutch navigators Pieter Dirkszoon Keyser and Frederick de Houtman.

Lacaille's star chart of the southern sky



1725

Flamsteed's atlas and catalog

John Flamsteed, England's first Astronomer Royal, produces the first major catalog of stars observed with the aid of a telescope. It is published posthumously along with *Atlas Coelestis*, and they become the standard references for the next century.

1751–52

More southern constellations

Nicolas Louis de Lacaille, a French astronomer, surveys the southern sky from the Cape of Good Hope, publishing a catalog of nearly 2,000 stars along with a star chart. He introduces 14 new southern constellations, all still recognized by astronomers.